

Polymers & Composites

Polymers

Recall that polymers consist of long carbon chains covalently bonded together, alongside side chains. Since the carbon chain is bonded together with single covalent bonds, they are free to rotate and bend, preventing close packing of the polymer chains and making the chains coil and entangle with other chains.

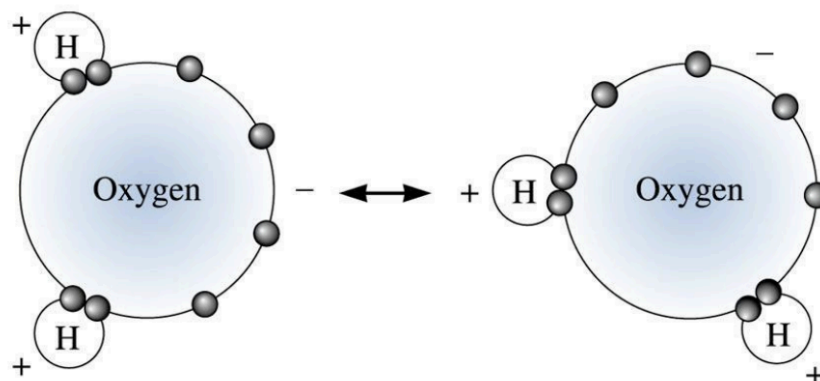
As the chains tangle, intermolecular forces can bond the tangled chains together.

Van Der Waals Bonding

The randomly moving electrons may sometimes cluster on one side of the atom, forming a temporary difference in electric charge, known as a dipole. The temporary dipole holds adjacent polymer chains together by their weak electrostatic attraction.

Hydrogen Bonding/Dipole-Dipole Forces

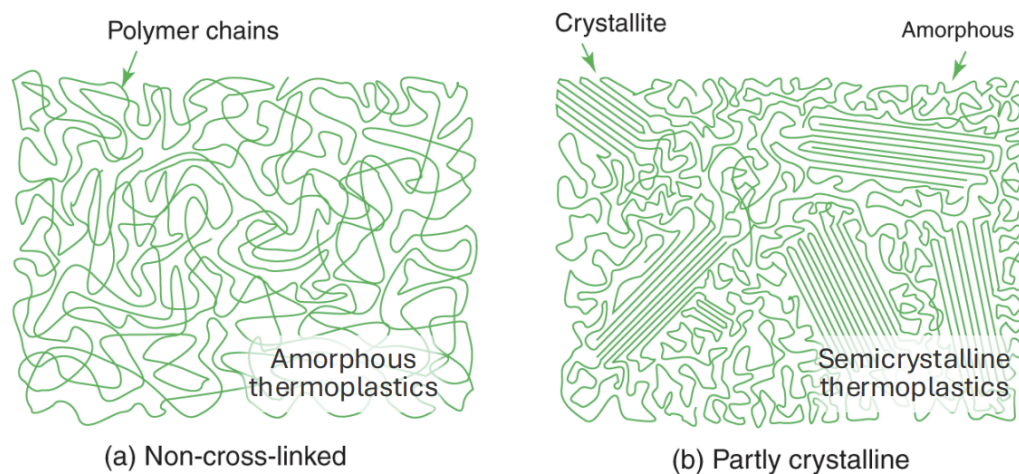
if a side group consists of hydrogen bonded with N, O or F, there is an imbalance in charge due to the bonding, forming a permanent dipole.



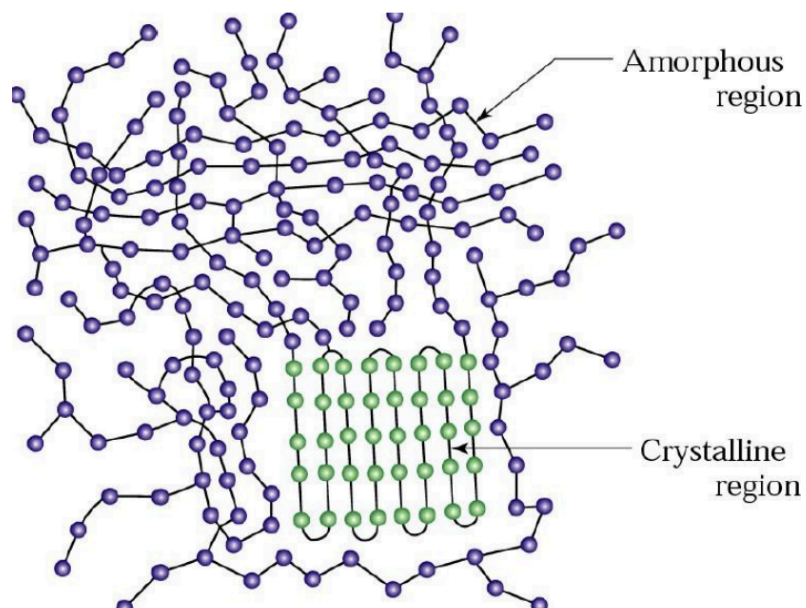
These form hydrogen bonds, which are 5 to 10 times stronger than Van Der Waals forces.

Other than hydrogen bonds, other permanent dipoles can form without hydrogen, like C-Cl bonds in PVC, which are also stronger than Van Der Waals forces.

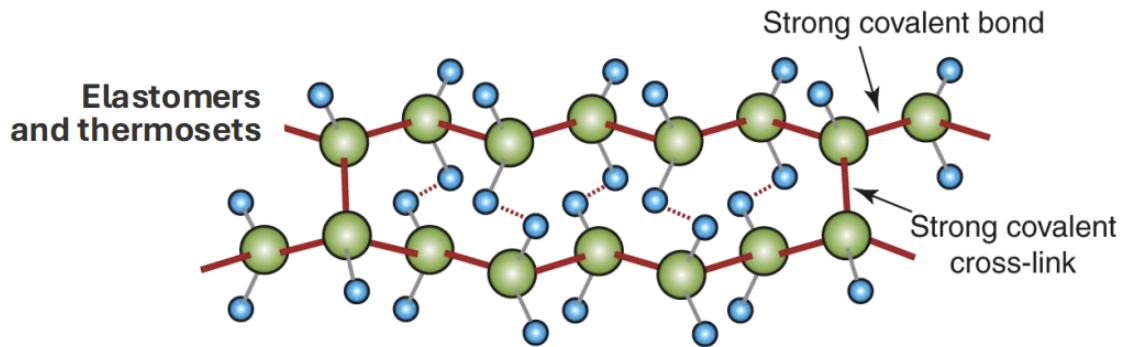
Because of these bonds, polymers like thermoplastics may sometimes form crystalline regions, due to the intermolecular forces/secondary bonding between strands. However, most thermoplastics usually have an amorphous arrangement.



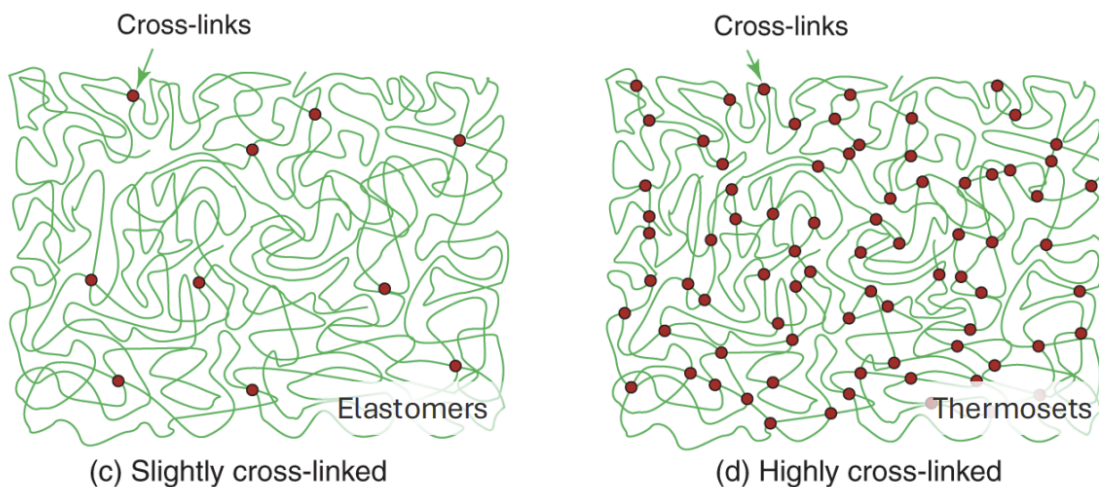
The amorphous regions (and amorphous thermoplastics) have no specific order, while the crystalline region has an ordered structure.



In the case of elastomers and thermosets, there are strong covalent bonds between the carbon atoms, forming cross links throughout the polymer, making them stiffer.



elastomers typically have lesser cross links compared to thermosets, making them slightly more flexible than thermosets:

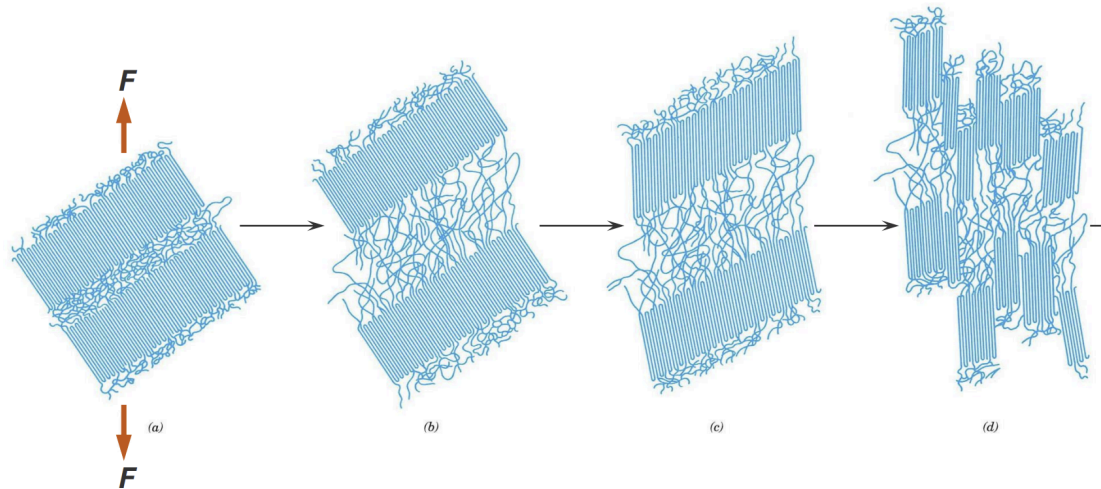


Thermoplastics

Under stress, the carbon backbone of polymers flex and stretch. In the case of amorphous regions/thermoplastics, the segments rotate about their bonds to uncoil and extend along the loading direction.

As further stress is applied, the polymer chains slide past one another, overcoming secondary bonds between strands.

For crystalline regions of thermoplastics, when a stress is applied, the crystalline regions separate from each other as the amorphous regions extend, until the crystalline regions are aligned in the direction of the stress, allowing the secondary bonds to be overcome by the stress, breaking apart the crystalline regions of the thermoplastic.



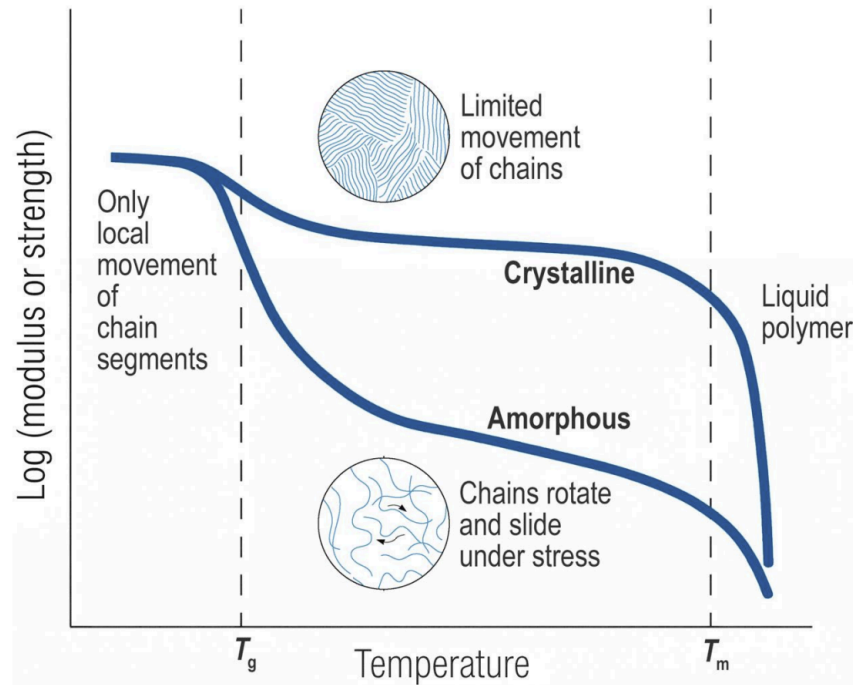
Structure-Property Relationships

Recall that hydrogen bonds are stronger than Van Der Waals forces. Hence, polymers with hydrogen bonds (or equivalent permanent dipole bonds) are stronger, stiffer and have higher yield strength. (e.g. nylon; polyamide functional group contains N-H, PVC; contains C-Cl bond) They also tend to have good toughness, unlike ceramics.

If there are more crystalline regions in the polymer (determined by the polymer processing), they resist deformation more, making them stiffer and have a higher yield strength, but reducing ductility (and hence toughness)

If the polymer chain has large side groups, like polypropylene and polystyrene (CH₃ and C₆H₅ side group respectively), they hinder chain uncoiling and sliding, increasing stiffness and strength but reducing ductility.

Thermoplastics also have a glass transition temperature $[T_g]$, and when above T_g , amorphous chains are able to slide under stress (different from chains just rotating locally about their bonds), decreasing their strength and stiffness, but improving their ductility. However, crystalline sections of thermoplastics are still unable to slide past one another fully, making their strength and stiffness relatively unchanged.



Hence, thermoplastics with glass transition temperature above room temperature makes them stiffer and stronger, compared to polymers with glass transition temperature at or below room temperature.

Thermosets

Due to the cross links in thermosets, chains are unable to slide past one another, but bonds can still flex and rotate. Hence, thermosets have good strength and stiffness, but low ductility.

Composites

Composites typically comprise of a matrix and a reinforcement, of which the reinforcement can be unidirectional, laminate, discontinuous chopped fibers, and particulate. Typically, a matrix is made out of a polymer, metal or ceramic.

The matrix holds and protects the reinforcement, and directs loads into the (typically) stronger reinforcement. The reinforcements provide the strength and stiffness of the composite itself, most effective with continuous fibers aligned in the direction of loading.

Hence, when a fiber reinforced composite is loaded along its fibers, or longitudinally, its stiffness is found by

$$E_c = E_f V_f + E_m V_m$$

where V_i is the volume fraction of i , i being the fiber or matrix. Hence, $V_i \in [0, 1]$ and $V_f + V_m = 1$.

If the composite is loaded perpendicularly, or transversely, the stiffness is found as

$$E_c = \frac{E_f E_m}{V_f E_m + V_m E_f}$$

Similar formulas can be used to find the strength of the composite, by replacing every E_i with σ_i .

Notice how $E_{c(longitudinal)} \geq E_{c(transverse)}$ for any combination of V_f and V_m , due to continuous fiber composites having anisotropy, or directional strength.

Hence, to reduce the anisotropy of composites, cross-ply laminates of continuous fibers are laid to ensure even strength, reducing failure in unexpected loadings.

The type of matrix used also determines its properties

Polymer matrix composites like CFRP tend to have high strength to weight ratios, and are corrosion and fatigue resistant, used in aircraft fuselage and other structures.

Ceramic matrix composites like SiC composites are able to withstand high temperatures due to ceramic's thermal stability, are lighter than typical metal alloys, and improve toughness of the ceramic, preventing failure when used as a lining in engines.

Metal matrix composites like Al matrix-SiC particulate composites, have higher stiffness and hardness due to the SiC, better thermal conductivity (and hence better heat distribution), and improves wear resistance, making it suitable to be used in brake discs.